RECEPTORS PHYSIOLOGY

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LEARNING OBJECTIVES

At the end of the lecture, students should be able to

- Define receptors
- Classify sensory receptors with examples
- Explain transduction of sensory stimuli in to nerve impulse
- Describe properties of sensory receptors
- Define receptor potential & generation of receptor potential.

DEFINITION

- Receptors are sensory (afferent) nerve endings that terminate in periphery as bare unmyelinated endings or in the form of specialized capsulated structures. **Sensory receptors** are specialized epithelial cells or neurons that transduce environmental signals in to neuronal signals.
- Receptors give response to the stimulus. When stimulated, receptors produce a series of impulses, which are transmitted through the afferent nerves.
- **Biological Transducers** Actually receptor functions like a transducer. So, receptors are often defined as the biological transducers, which convert (transducer) various forms of energy (stimuli) in the environment into action potentials in nerve fiber.

TYPES OF RECEPTORS
BASED ON LOCATION

- **EXTERORECEPTORS**: Receptors which give response to stimuli arising from outside the body i.e. associated with skin
- **VISCERORECEPTORS**: Receptors situated in the viscera i.e. associated with organs
- **PROPRIOCEPTORS**: Receptors which give response to change in the position of different parts of the body. associated with joints, tendons

TYPES OF SENSORY RECEPTORS
According To Types Of Stimulus

- **Mechanoreceptors**: Receptors situated in the skin are called the cutaneous receptors. Cutaneous receptors are also called mechanoreceptors because of their response to mechanical stimuli such as touch, pressure and pain. Touch and pressure receptors give response to vibration also. Eg.; compression, bending, stretching of cells. Touch, pressure, proprioception, hearing, and balance
- **Chemoreceptors**: Receptors, which give response to chemical stimuli, are called the chemoreceptors. Chemicals become attached to receptors on their membranes. Eg; Smell and taste
- **Thermoreceptors**: respond to changes in temperature
- **Photoreceptors**: respond to light: vision
- **Nociceptors**: extreme mechanical, chemical, or thermal stimuli.Eg; Pain

CUTANEOUS RECEPTORS

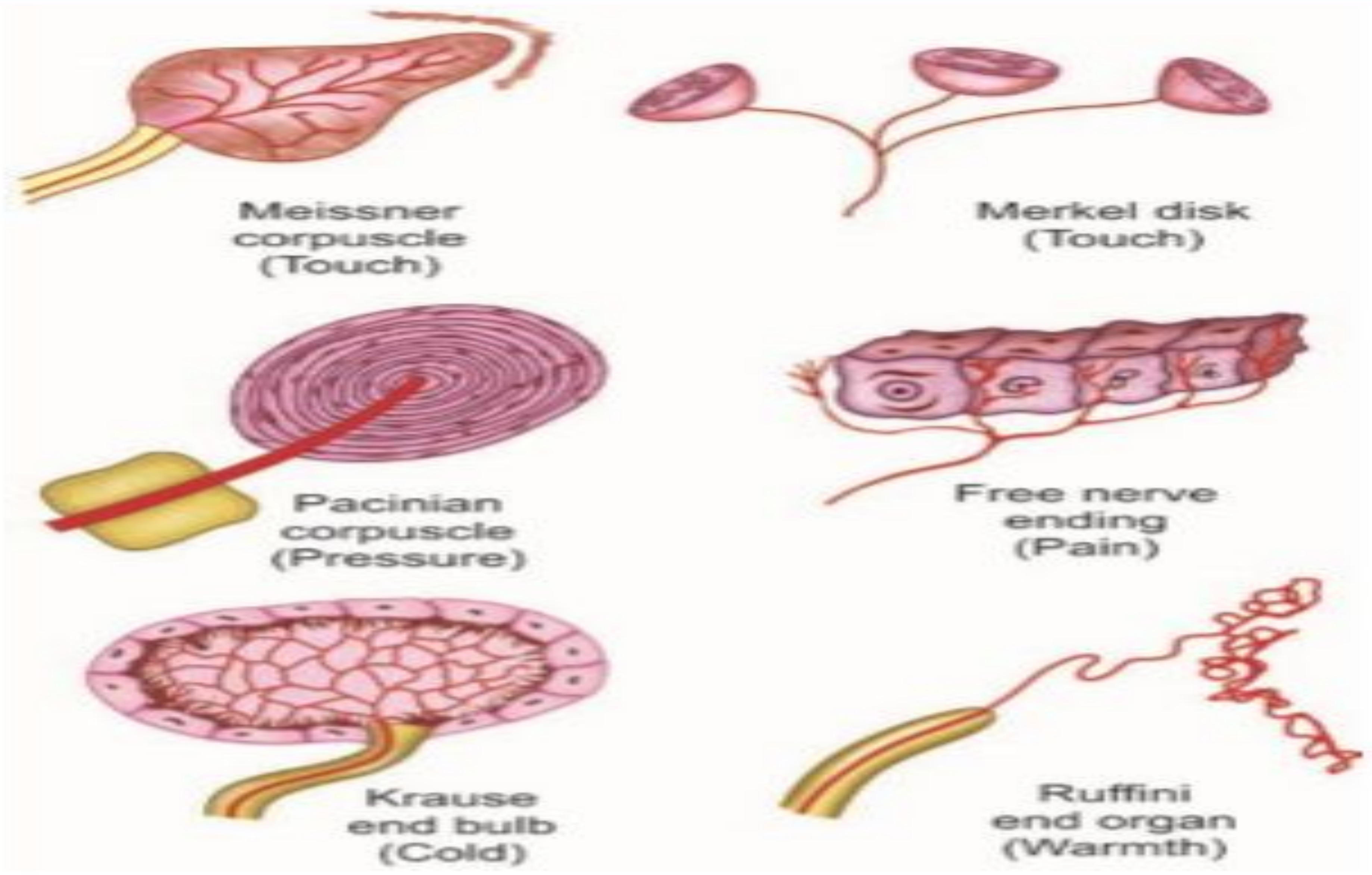


Table 3-2 Types and Examples of Sensory Receptors

Type of Receptor	Modality	Receptor	Location
Mechanoreceptors	Touch	Pacinian corpuscle	Skin
	Audition	Hair cell	Organ of Corti
	Vestibular	Hair cell	Macula, semicircular canal
Photoreceptors	Vision	Rods and cones	Retina
Chemoreceptors	Olfaction	Olfactory receptor	Olfactory mucosa
	Taste	Taste buds	Tongue
	Arterial Po_2		Carotid and aortic bodies
	pH of CSF		Ventrolateral medulla
Thermoreceptors	Temperature	Cold receptors	Skin
		Warm receptors	Skin
Nociceptors	Extremes of pain and temperature	Thermal nociceptors	Skin
		Polymodal nociceptors	Skin

CSF, Cerebrospinal fluid; Po_2, partial pressure of oxygen.

CHEMORECEPTORS

- **Olfactory receptors** -- Sensations of smell.

Detect *air borne chemicals,* relay an electrical impulse via axons of chemoreceptor cells to olfactory bulb

- **Gustatory receptors** -- Sensations of taste.

Detection of *chemicals in a solution*, relay an electrical impulse via sensory neurons to **thalamus** for sorting, sends on to appropriate higher brain center.

EXTEROCEPTORS

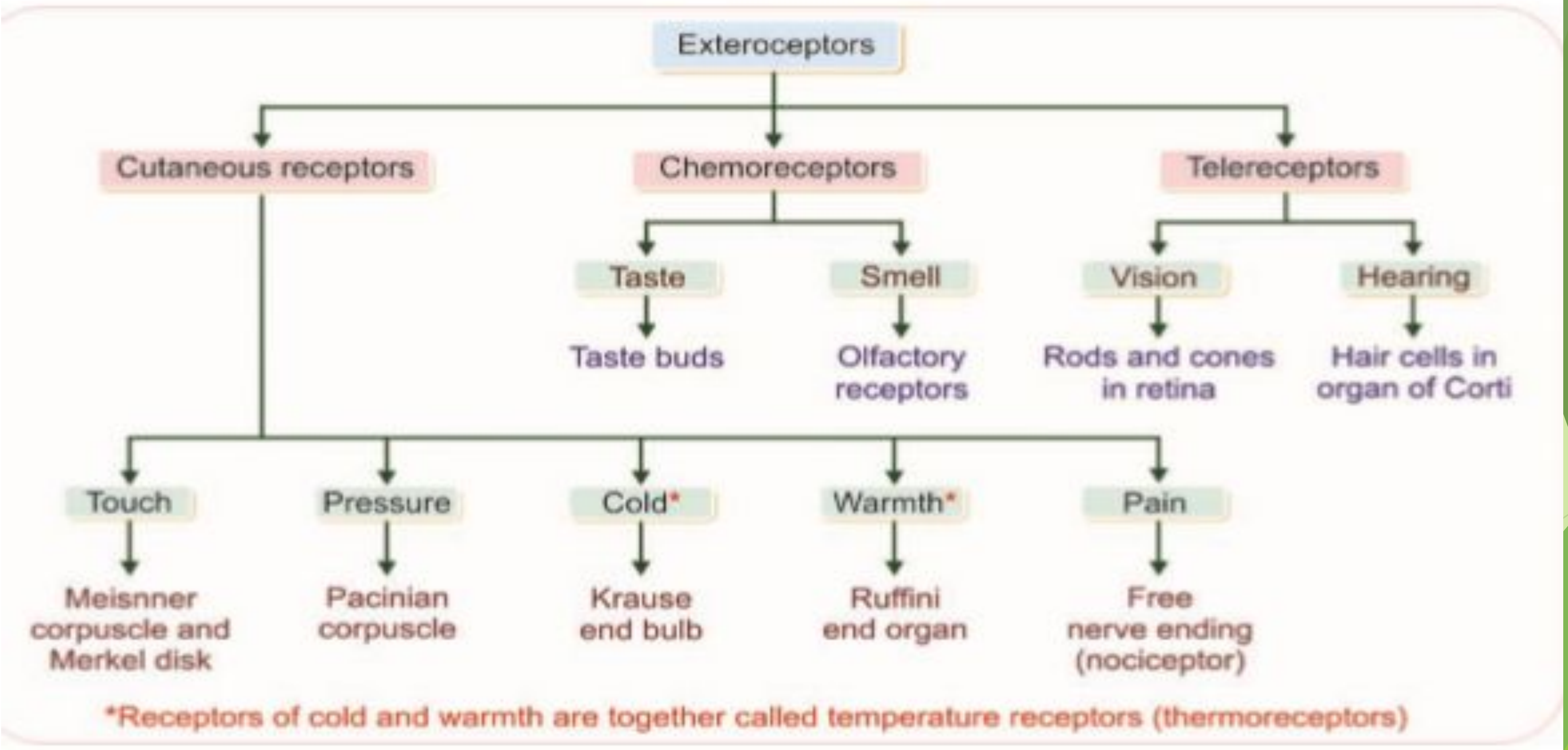
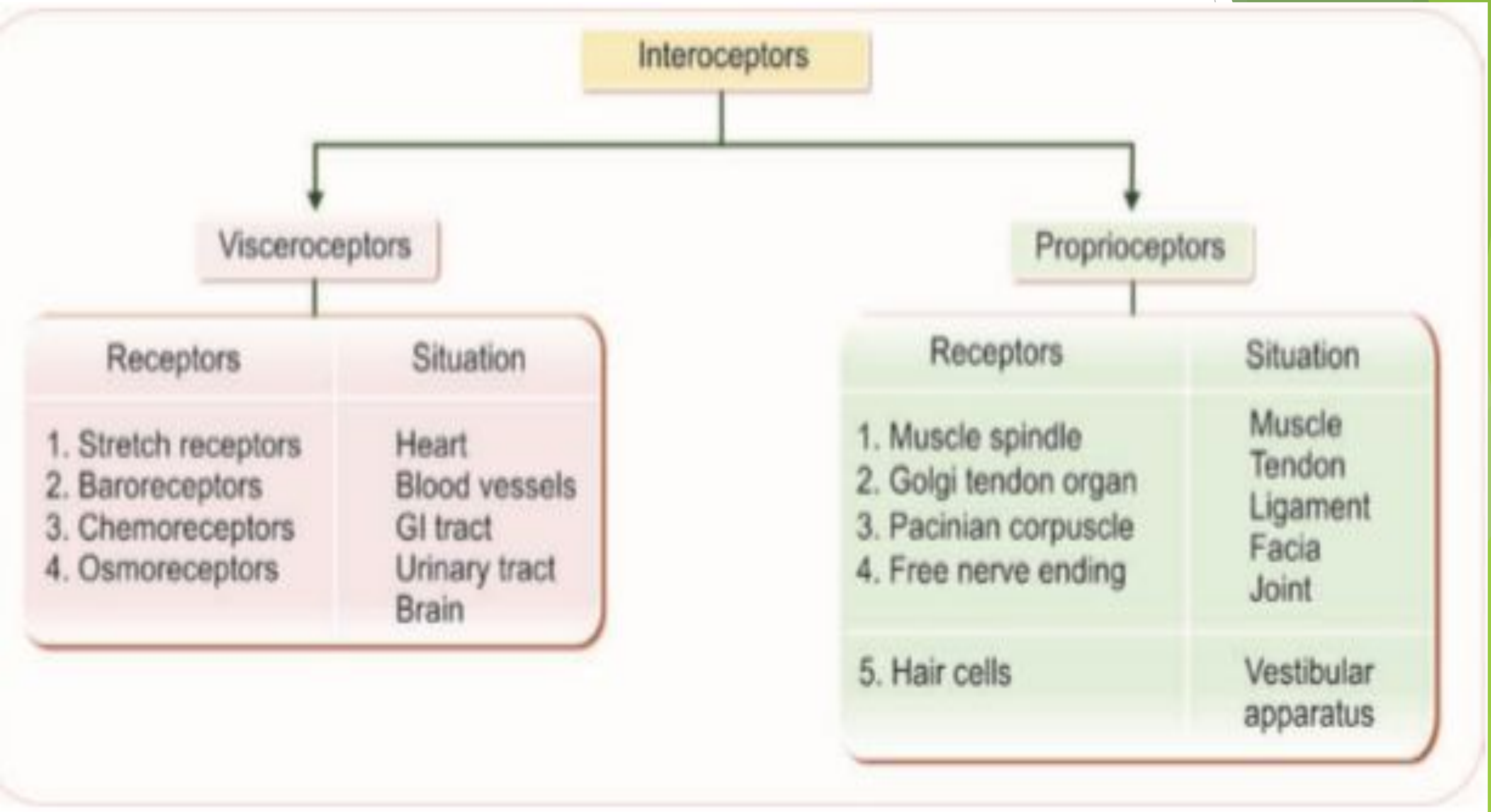


FIGURE 139.2: Exteroceptors

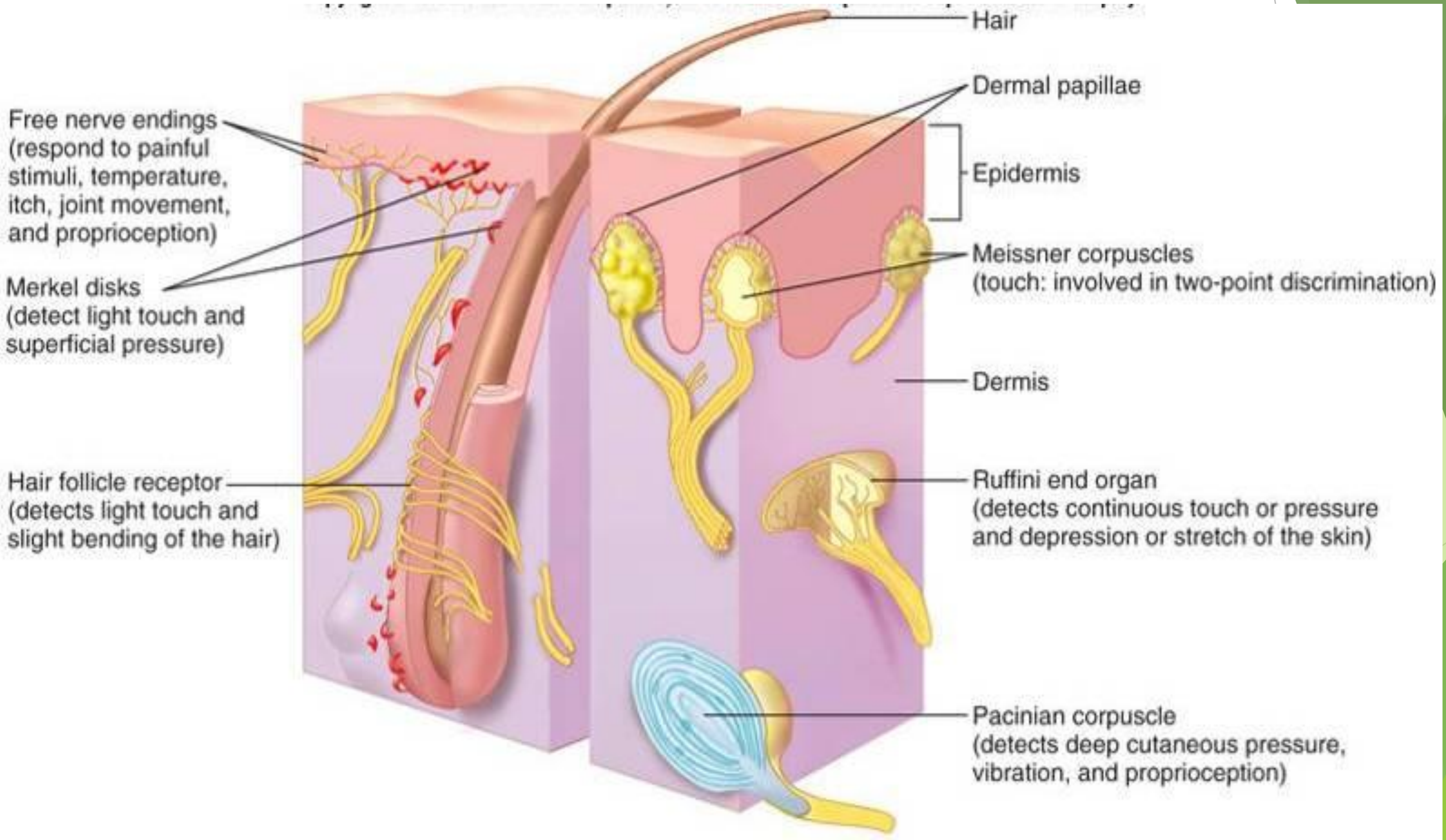
INTEROCEPTORS



FREE NERVE ENDINGS

- Simplest, most common sensory receptor
- Scattered through most of body; visceroceptors are of this type.
- Type responsible for temperature sensation
  - Cold: 10-15 times more numerous than warm
  - Warm
  - Pain: responds to extreme cold or heat

SENSORY NERVE ENDINGS IN SKIN



MERKEL (TACTILE) DISKS

- Axonal branches end as flattened expansions associated with epithelial cells
- Basal layers of epidermis
- Associated with dome-shaped mounds of thickened epidermis in hairy skin
- Light touch and superficial pressure.

HAIR FOLLICLE RECEPTORS

- Hair end organs
- Respond to slight bending of hair as occurs in light touch
- End organ receptor fields overlap; sensation not very localized, yet very sensitive

PACINIAN CORPUSCLES

- Lamellated corpuscles
- Single dendrite to layers of corpuscles arranged like leaves of an onion
- Deep dermis or hypodermis
- Deep cutaneous pressure; vibration
- When associated with joints, involved in proprioception

MEISSNER (TACTILE) CORPUSCLES

- Present in non hairy skin.
- Two-point discrimination
- Ability to detect simultaneous stimulations at two points on the skin.
- Used to determined texture of objects.
- Numerous and close together on tongue and fingertips

TWO POINT DISCRIMINATION

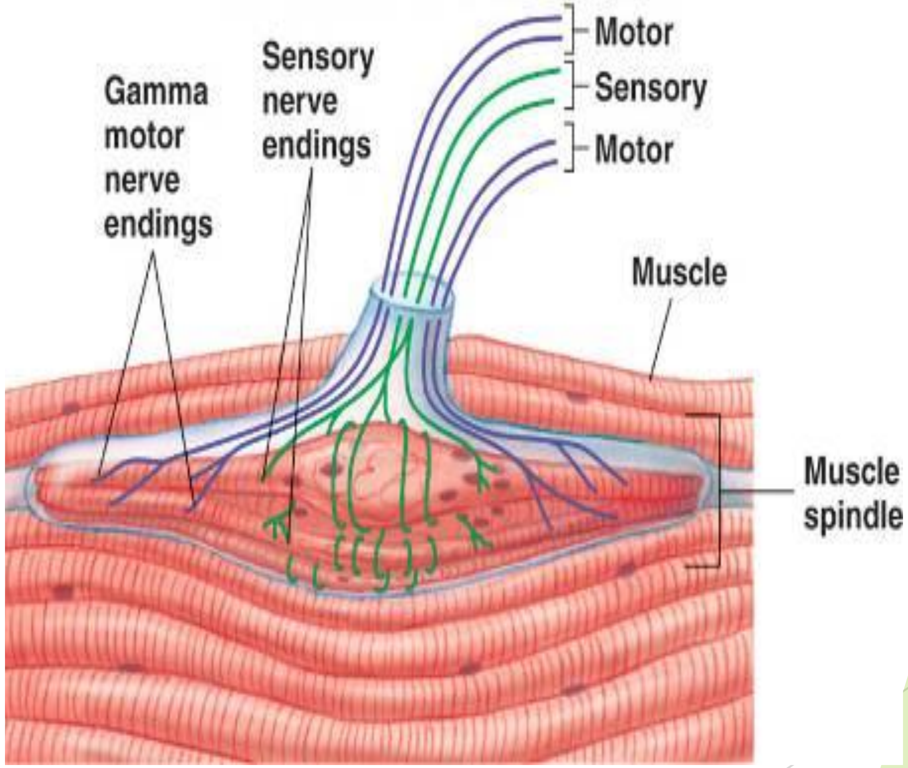
- Decreased overlap
- Increased discrimination
- Highly accurate on tips of finger, lips, tongue.

RUFFINI END ORGAN

- Primarily in dermis of fingers
- Respond to continuous touch or pressure

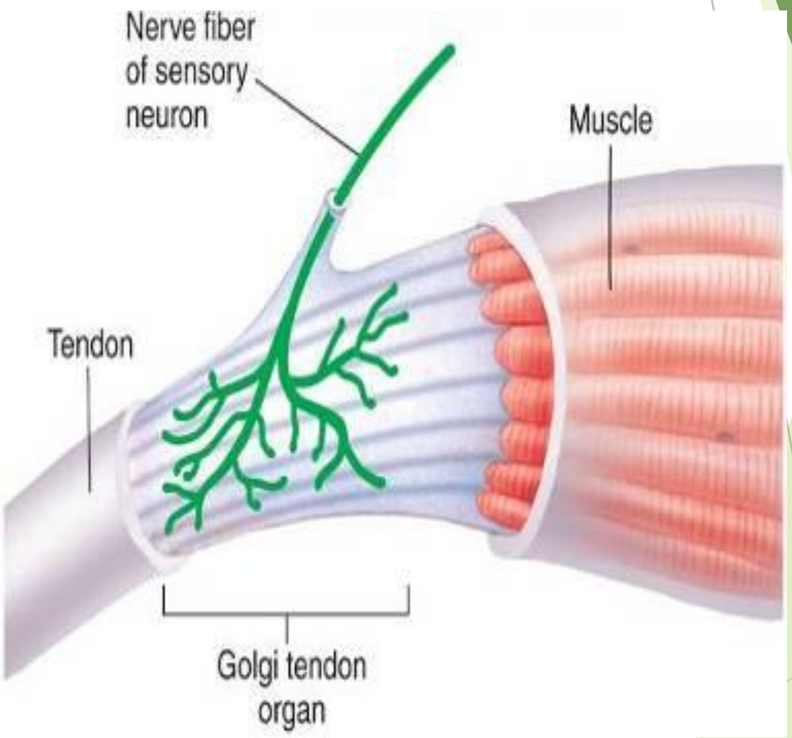
MUSCLE SPINDLES

- **3-10 specialized skeletal muscle cells**
- **Provide information about length of muscles**
- **Involved in stretch reflex**



GOLGI TENDON ORGAN

- Proprioceptors associated with tendons
- Respond to increased tension on tendon



RESPONSES OF SENSORY RECEPTORS

- Graded potential: results from interaction of sensory receptor with stimulus
- Graded potential is called a receptor or generator potential
  - Primary receptors: axons conduct action potentials in response to receptor potential
  - Secondary receptors: cause release of neurotransmitters that bind to receptors on a neuron causing a receptor potential. Smell, taste, hearing, balance

Sensory Transduction and Receptor Potentials

- Sensory transduction is the process by which an environmental stimulus (e.g., pressure, light, chemicals) activates a receptor and is converted into electrical energy.
- The conversion typically involves opening or closing of ion channels in the receptor membrane, which leads to a flow of ions (current flow) across the membrane.
- Current flow then leads to a change in membrane potential, called a receptor potential, which increases or decreases the likelihood that action potentials will occur.

The following series of steps occurs when a stimulus activates a sensory receptor:

- The environmental **stimulus** interacts with the sensory receptor and causes a change in its properties

- A mechanical stimulus causes *movement* of the mechanoreceptor (e.g., sound waves move the hair cells in the organ of Corti)

- Photons of light are absorbed by pigments in photoreceptors on the retina, causing photoisomerization of rhodopsin (a chemical in the photoreceptor membrane)

- In each case, a change occurs in the sensory receptor.

The following series of steps occurs when a stimulus activates a sensory receptor:

- These changes cause ion channels in the sensory receptor membrane to open or close, which results in a change in current flow
- If ionic current flow is inward (i.e., positive charges move into the receptor cell), then depolarization occurs
- If current flow is outward (i.e., positive charges move out of the cell), then hyperpolarization occurs
- The resulting change in membrane potential, either depolarization or hyperpolarization, is called the **receptor potentialor generator potential**
- The receptor potential is not an action potential. Rather, the receptor potential increases or decreases the likelihood that an action potential will occur, depending on whether it is depolarizing or hyperpolarizing
- Receptor potentials are graded electronic potentials, whose amplitude correlates with the size of the stimulus.

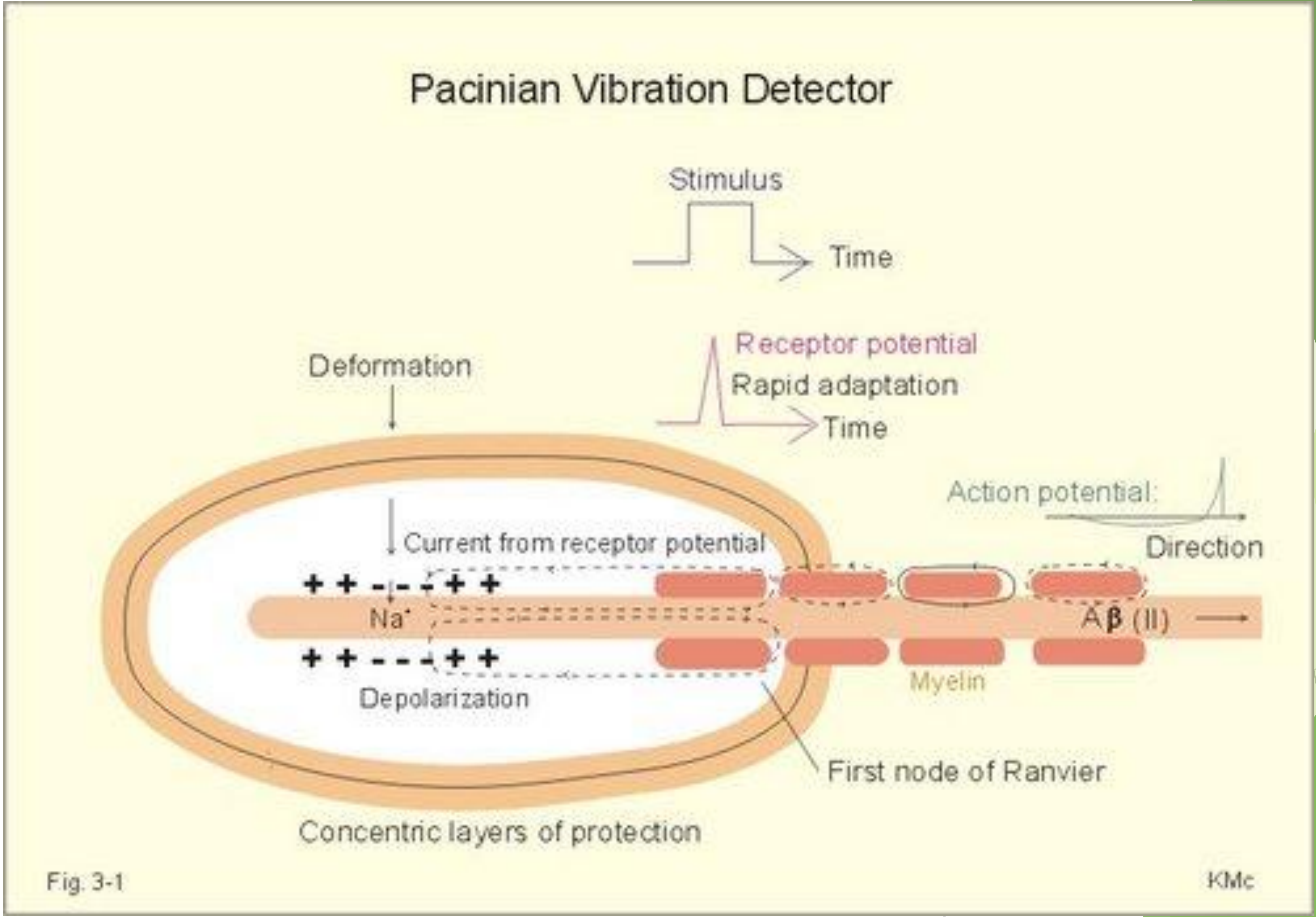
The following series of steps occurs when a stimulus activates a sensory receptor:

- If the receptor potential is depolarizing, it moves the membrane potential toward threshold and increases the likelihood that an action potential will occur

- Because receptor potentials are graded in amplitude, a small depolarizing receptor potential still may be subthreshold and, therefore, insufficient to produce an action potential

- However, a larger stimulus will produce a larger depolarizing receptor potential, and if it reaches or exceeds threshold, action potentials will occur

- If the receptor potential is hyperpolarizing , it moves the membrane potential away from threshold, always decreasing the likelihood that action potentials will occur.


Pacinian Vibration Detector
Stimulus
Time
Receptor potential
Rapid adaptation
Time
Deformation
Action potential:
Direction
Current from receptor potential
+ + - - - + +
Na+
Aβ (II)
+ + - - - + +
Depolarization
Myelin
First node of Ranvier
Concentric layers of protection
Fig. 3-1
KMc

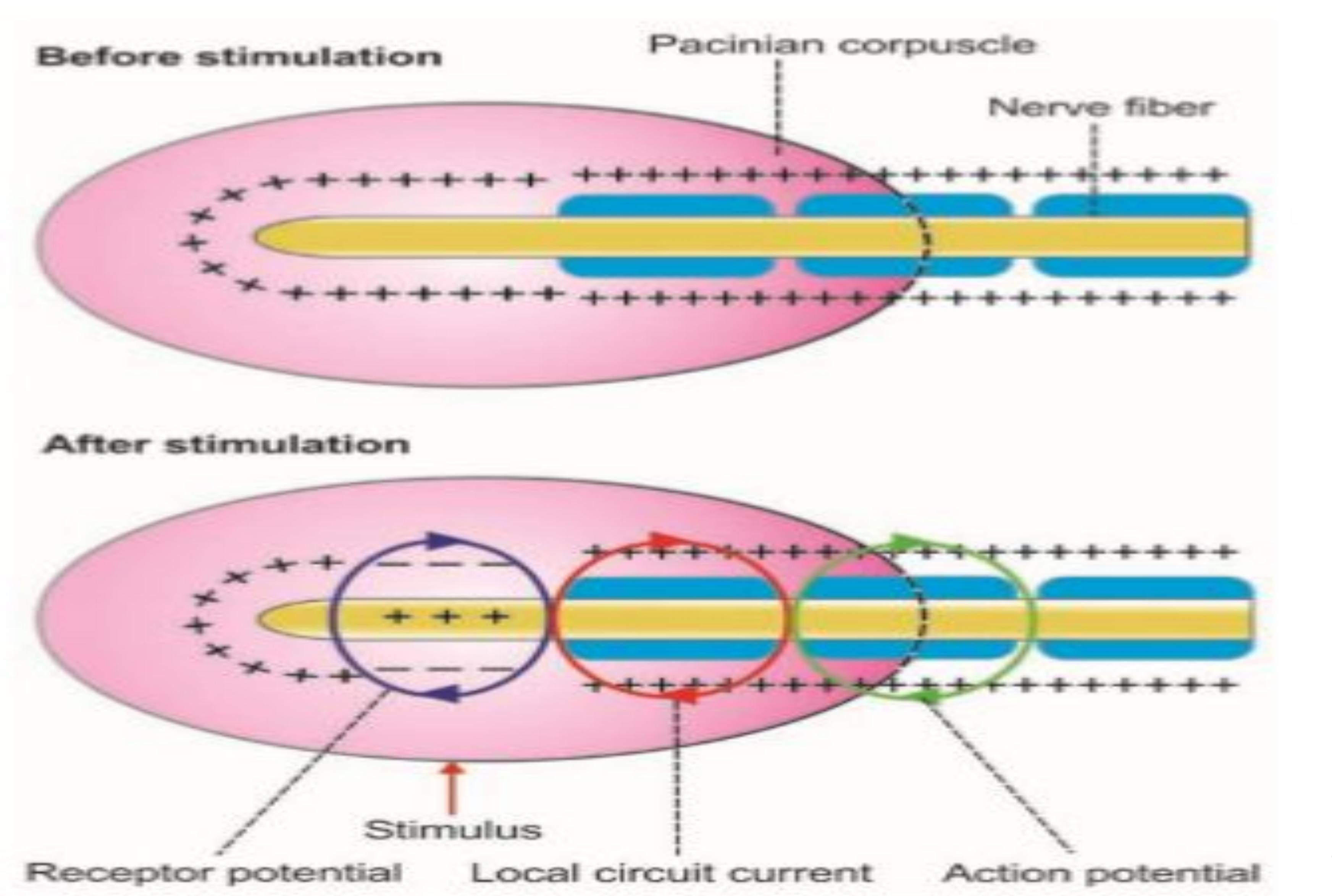


FIGURE 139.4: Receptor potential in pacinian corpuscle. Receptor potential leads to development of local circuit which spreads up to first node within the capsule. It leads to development of action potential in the first node of nerve fiber.

PROPERTIES OF RECEPTORS:

1. SPECIFICITY OF RESPONSE OR MULLER LAW

- **Specificity of response or Müller law** refers to the response given by a particular type of receptor to a specific sensation. For example, pain receptors give response only to pain sensation. Similarly, temperature receptors give response only to temperature sensation. In addition, each type of sensation depends upon the part of the brain in which its fibers terminate. Specificity of response is also called Müller's doctrine of specific nerve energies.

2. SENSORY ADAPTATION

- **Adaptation** is the decline in discharge of sensory impulses when a receptor is stimulated continuously with constant strength. It is also called sensory adaptation or desensitization.
- Depending upon adaptation time, receptors are divided into two types:

i. Phasic receptors, which get adapted rapidly. Touch and pressure receptors are the phasic receptors

ii. Tonic receptors, which adapt slowly. Muscle spindle, pain receptors and cold receptors are the tonic receptors.

3. RESPONSE TO INCREASE IN STRENGTH OF STIMULUS – WEBERFECHNER LAW

- During the stimulation of a receptor, if the response given by the receptor is to be doubled, the strength of stimulus must be increased 100 times. This phenomenon is called Weber Fechner law, which states that intensity of response (sensation) of a receptor is directly proportional to logarithmic increase in the intensity of stimulus.

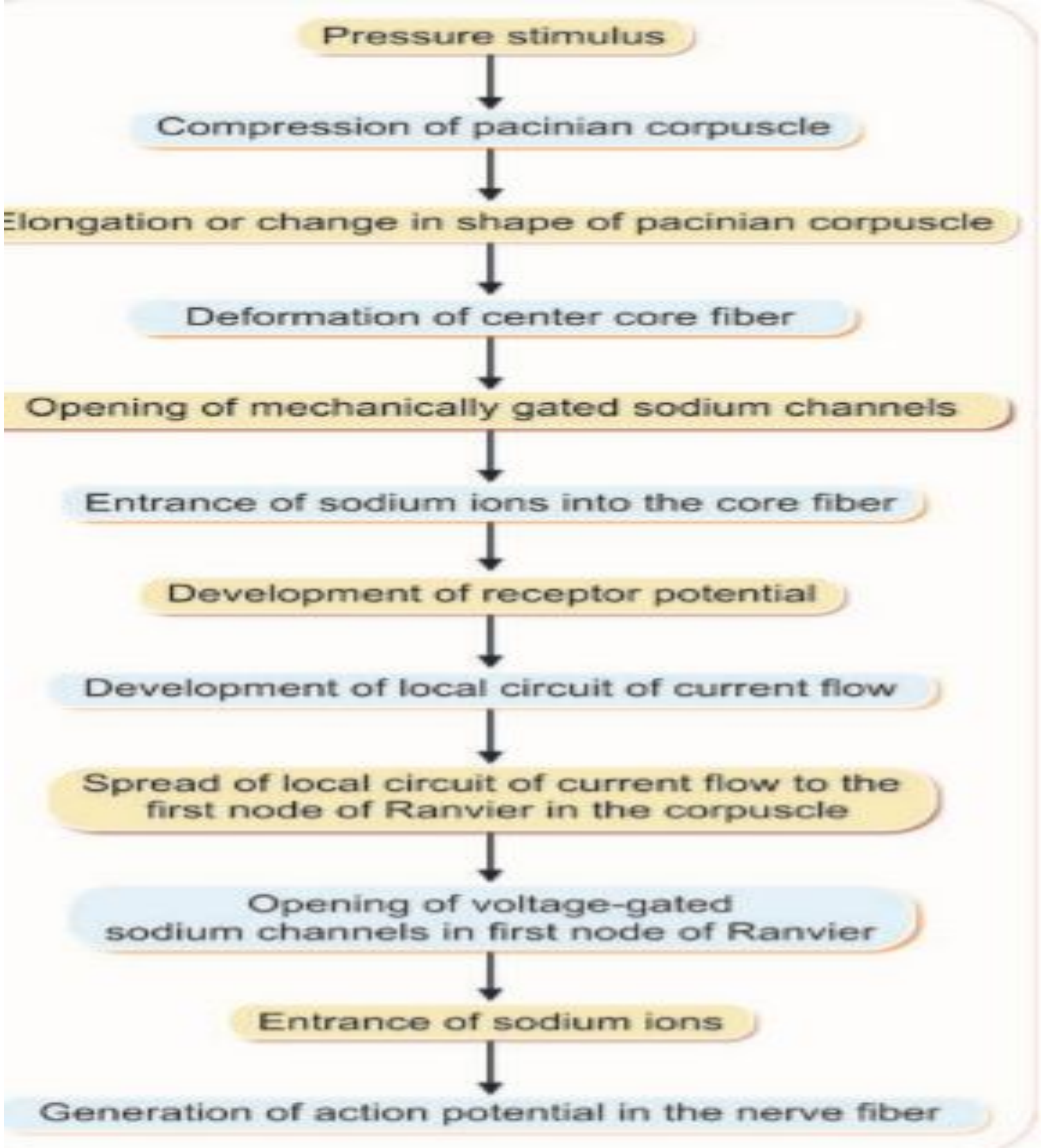
4. SENSORY TRANSDUCTION

- Sensory transduction in a receptor is a process by which the energy (stimulus) in the environment is converted into electrical impulses (action potentials) in nerve fiber (transduction = conversion of one form of energy into another).
- When a receptor is stimulated, it gives response by sending information about the stimulus to CNS. Series of events occur to carry out this function such as the development of receptor potential in the receptor cell and development of action potential in the sensory nerve.
- Sensory transduction varies depending upon the type of receptor. For example, the chemoreceptor converts chemical energy into action potential in the sensory nerve fiber. Touch receptor converts mechanical energy into action potential in the sensory nerve fiber.

5. RECEPTOR POTENTIAL

- Receptor potential is a nonpropagated transmembrane potential difference that develops when a receptor is stimulated. It is also called generator potential. Receptor potential is short lived and hence, it is called transient receptor potential.
- Receptor potential is not action potential. It is a graded potential. It is similar to excitatory postsynaptic potential (EPSP) in synapse, endplate potential in neuromuscular junction and electrotonic potential in the nerve fiber.
- **Properties of Receptor Potential** Receptor potential has two important properties. i. Receptor potential is nonpropagated; it is confined within the receptor itself ii. It does not obey all or none law.
- **Significance of Receptor Potential** When receptor potential is sufficiently strong (when the magnitude is about 10 mV), it causes development of action potential in the sensory nerve.

Schematic diagram showing development of receptor potential and generation of action potential in nerve fiber.



6. LAW OF PROJECTION

- When a sensory pathway from receptor to cerebral cortex is stimulated on any particular site along its course, the sensation caused by stimulus is always felt (referred) at the location of receptor, irrespective of site stimulated. This phenomenon is known as law of projection.
- Eg. Sensation complained by amputated patients in the missing limb (phantom limb) is the best example of law of projection. For example, if a leg has been amputated, the cut end heals with scar formation. The cut ends of nerve fibers are merged within the scar. If the cut end of sensory fibers are stimulated during movement of thigh, the patient feels as if the sensation is originating from nonexistent leg. Sometimes, the patient feels pain in nonexistent limb. This type of pain is called phantom limb pain.

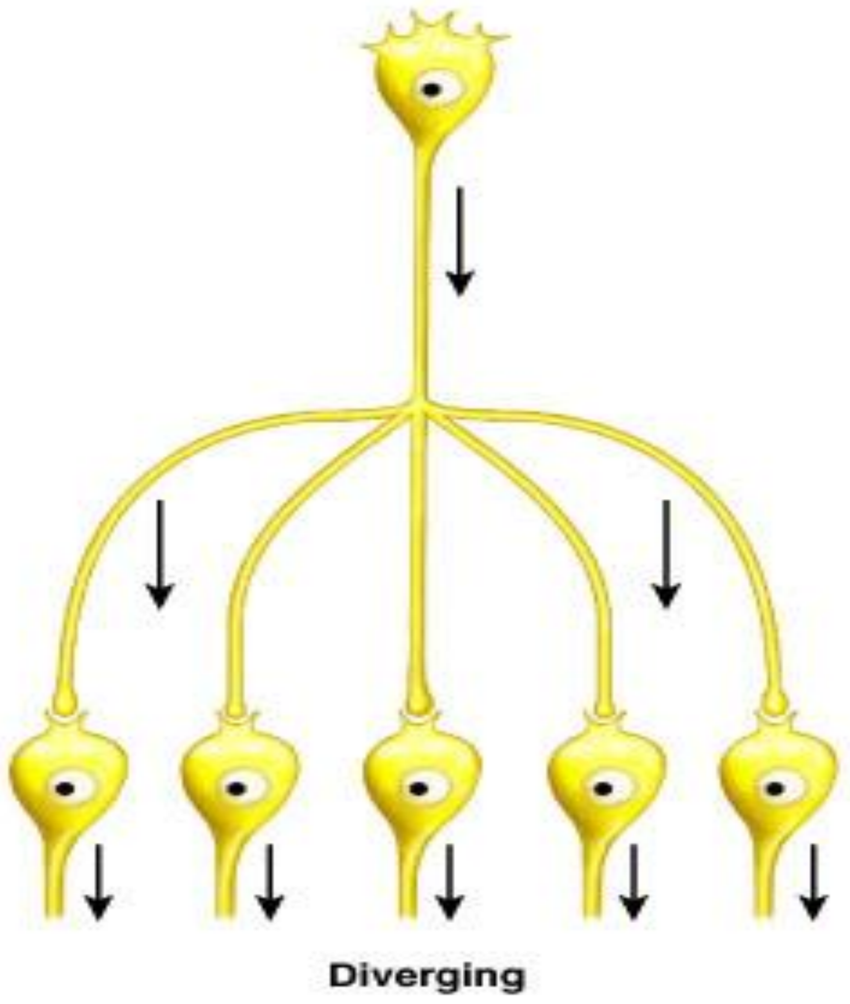
NEURONAL CIRCUITS

NEURONAL CIRCUITS

- Diverging circuit
- Converging circuit
- Reverberating circuit

DIVERGING CIRCUIT

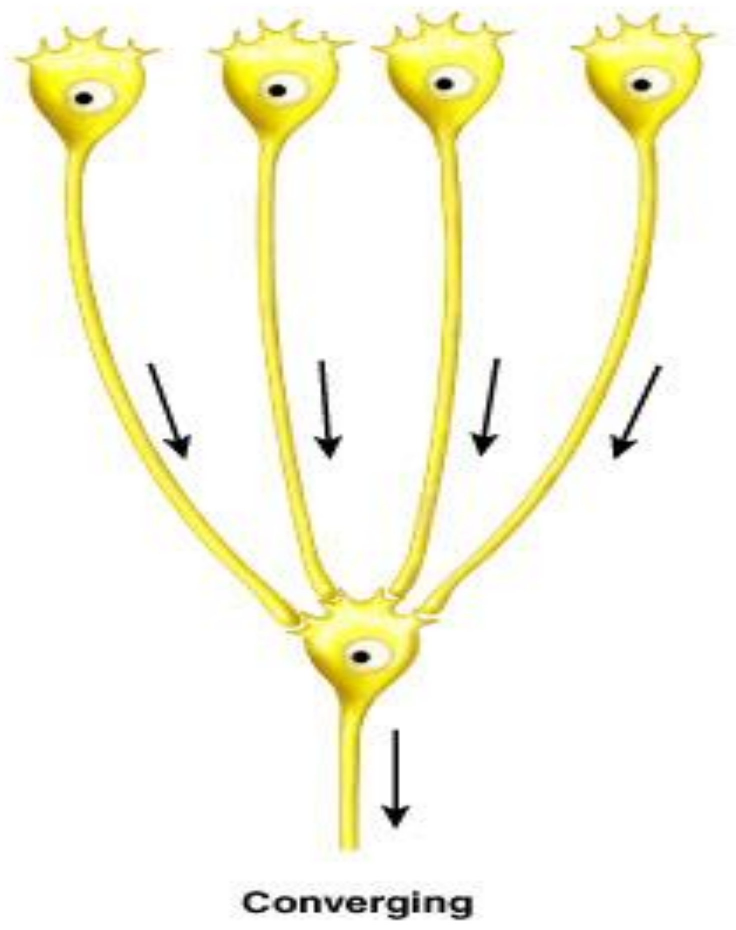
- Diverging circuit – one presynaptic neuron synapses with several other neurons (divergence)



(a) Types of neuronal circuits

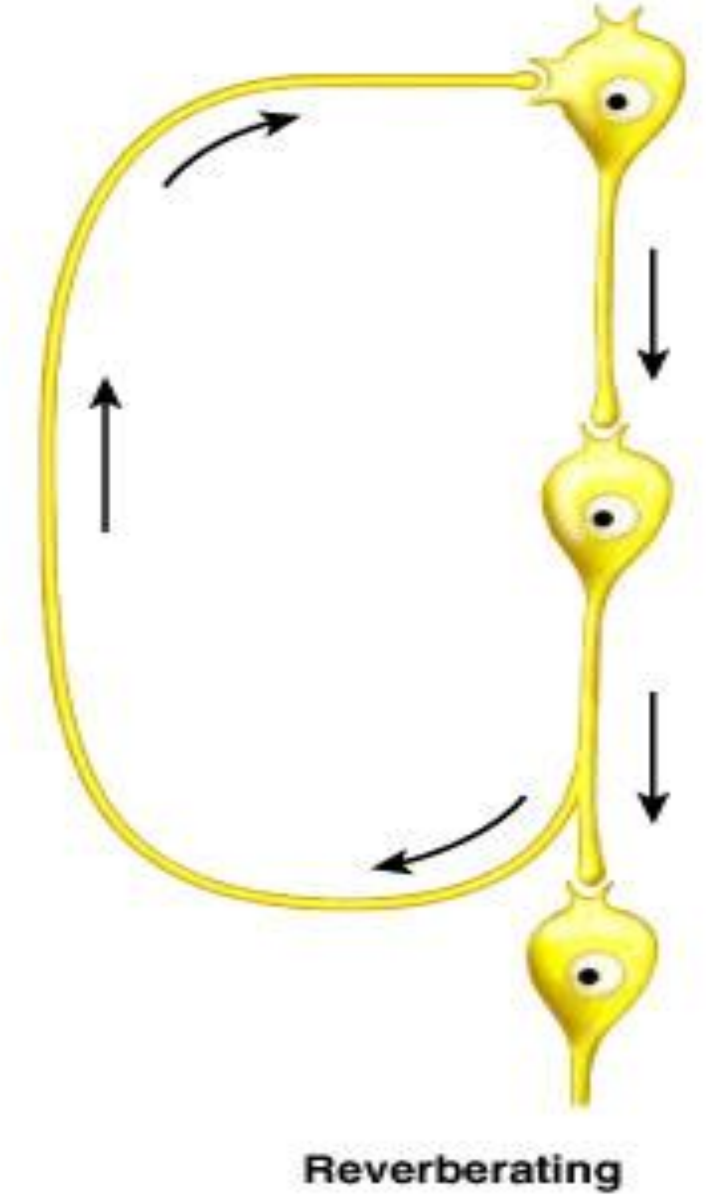
CONVERGING CIRCUIT

- Converging circuit – many neurons synapse on a single postsynaptic neuron (convergence)



REVERBERATING CIRCUIT

- Reverberating circuit – circuit that receives feedback via a collateral axon from a neuron in the circuit



NEURAL PROCESSING

1: Serial processing

2: Parallel processing

SERIAL PROCESSING

- Serial processing – neurons pass a signal to a specific destination along a single pathway from one to another

Serial processing

(b) Types of neural processing

PARALLEL PROCESSING

- Parallel processing – input is delivered along many pathways; a single sensory stimulus results in multiple perceptions